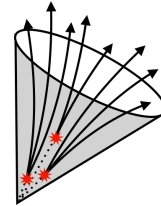


Unsupervised Anomaly Detection of Emerging Jets at the LHC using a model-agnostic track-based Machine Learning method

Project Description:

A compelling class of dark-matter theories proposes the existence of a dark sector, potentially endowed with its own confining gauge interactions, known as “dark QCD”. If produced at the LHC, dark quarks may undergo a hidden-sector shower and hadronisation, yielding long-lived dark hadrons that eventually decay back into Standard Model (SM) particles.

These dynamics give rise to non-standard jet signatures, among which Emerging Jets (EJs) are particularly striking. EJs are jets that initially consist of invisible long-lived dark particles, which travel some distance inside the detector before decaying into detectable SM particles. As these delayed decays occur, visible tracks and hadronic activity suddenly appear, giving rise to a jet that seems to “emerge” within the detector volume. They are characterised by numerous tracks that do not point back to the interaction point, along with multiple displaced vertices. However, their properties depend sensitively on unknown dark-sector coupling parameters and mass scales, supervised methods to perform poorly across the vast and largely unexplored parameter space.



In the ATLAS experiment at the LHC, such displaced activity can be probed through track-level observables in the inner detector, such as impact parameters and hit-pattern information. Unlike supervised classifiers, which require explicit labels for exotic signatures, unsupervised models are well-suited for dark-sector scenarios where lifetimes and mass hierarchies are unknown and no reliable truth-level labelling exists.

Identifying such jets is a major challenge at collider experiments due to:

- the complex QCD environment,
- the subtle and gradual onset of displaced tracks,
- the lack of reliable supervised labels for exotic signatures.

This project aims to explore whether advanced unsupervised machine-learning models trained exclusively on SM QCD jets can detect such dark-sector-induced anomalies. Using rich track-level information the goal is to learn the structure of ordinary QCD jets and identify deviations compatible with a variety of dark-QCD EJ scenarios.

This project aims to develop an **unsupervised anomaly-detection pipeline** trained solely on SM QCD jets. The goal is to determine whether advanced ML architectures, using track-level information can learn the natural structure of QCD jets and identify as anomalous those jets containing EJ-like patterns.

The project lies at the intersection of **experimental particle physics**, jet substructure, and **modern ML for anomaly detection**, and is ideal for students interested in collider physics or frontier applications of deep learning.

Tasks and Expected Outcome:

Task 1 — Data Exploration and Preprocessing

The dataset is inspired by jet reconstruction in the ATLAS detector at the LHC, where track-level information from the inner detector plays a key role in identifying displaced activity.

The student is expected to:

- Inspect jet-level and track-level features.
- Perform feature cleaning, normalisation, and masking of variable-length track sets.
- Implement different representations: fixed-size embeddings, padded sequences, sets/graphs.

Task 2 — Implementation of Unsupervised Models

Students will explore state-of-the-art unsupervised models such as:

- Autoencoders / Variational Autoencoders
- Normalising Flows
- Diffusion Models
- Graph Neural Networks (GNNs) on jet constituents
- Transformer architectures for unordered sets of tracks

This step includes:

- model design and training on SM **QCD-only simulated data**,
- tuning latent dimensions, regularisation, input encodings,
- anomaly-score definition (reconstruction error, likelihood, flow score, etc.).

Task 3 — Evaluation on Mixed QCD+EJ Samples

- Compute anomaly-score distributions for QCD vs. EJ.
- Evaluate performance via:
 - ROC curves and AUC,
 - efficiency vs. mistag rate curves,
 - sensitivity trends as a function of LLP decay length and mass.
- Performance comparison with a supervised approach
- Study model interpretability (optional): feature attribution or latent-space visualisation.

Expected Outcome

By the end of the project, students will deliver:

- A functioning pipeline for unsupervised anomaly detection on hadronic jets.
- Comparative performance metrics (ROC/AUC, score distributions).
- A demonstration that track-level patterns increase sensitivity to displaced activity.
- A concise report and final presentation summarising methodology, ML design choices, results, and physics interpretation.

Available Material:

Dataset of QCD and EJ-like events, including:

- jet-level variables,
- full track-level features (impact parameters, hit patterns, momenta, uncertainties).

Access to Leonardo HPC at CINECA through course-provided credentials.

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Bibliographic References:

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2. *ATLAS Collaboration, Search for emerging jets in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS experiment, Rept. Prog. Phys. 88 (2025) 9, 097801*